TYPE CASTING IN JAVA

🔄 Type Casting in Java (Simple Explanation)

Today I learned about Type Casting in Java.

What is Type Casting?

Type Casting means converting one data type into another data type.

Sometimes in programming we need to store a value of one type into another type variable.
To do this, Java allows type conversion, which is called Type Casting.

Example:
Convert int → double
Convert double → int

◇ Why Type Casting is Needed?

Type casting is used when:

• We want to store a value in a different data type
• We want to perform calculations with different types
• We want to reduce memory usage
• We want precise control over data conversion

◇ Types of Type Casting in Java

Java has two types of Type Casting:

1. Widening Casting (Implicit Casting)
2. Narrowing Casting (Explicit Casting)

1. Widening Casting (Implicit casting)

Widening casting happens when a smaller data type is converted into a larger data type automatically.

📌 Java does this automatically, so we do not need to write any special code.

Example Conversion Order

byte → short → int → long → float → double

💻 **Example Code :**

```
public class WideningCasting {
   public static void main(String[] args) {

      int number = 25;

      double result = number;  // automatic conversion

      System.out.println(number);
      System.out.println(result);
   }
}
```

📌 **Output**

```
25
25.0
```

Important Points

- Done automatically by Java
- No data loss
- Safe conversion
- Smaller → Bigger data type

2.Narrowing Casting (Manual)

Narrowing casting happens when a larger data type is converted into a smaller data type.

Java cannot do this automatically, so we must write the data type in brackets.

Conversion Order

double → float → long → int → short → byte

💻 Example Code

```
public class NarrowingCasting {
   public static void main(String[] args) {

       double number = 19.99;

       int result = (int) number;  // manual casting

       System.out.println(number);
       System.out.println(result);
   }
}
```

📌 Output

```
19.99
19
```

Important Points

• Must be written manually
• May cause data loss
• Decimal values may be removed

Example:

10.75 → 10

Difference Between Implicit and Explicit Type Casting in Java

Feature	Implicit Type Casting	Explicit Type Casting
Another Name	Widening Casting	Narrowing Casting
Conversion	Smaller → Larger data type	Larger → Smaller data type
Done By	Java automatically	Programmer manually
Syntax Needed	Not required	Required (type)
Data Loss	No data loss	Possible data loss
Safety	Safe conversion	Risky conversion

◇ Real Life Example

Imagine a big water bottle → small glass

You cannot fit all water in the glass.

Some water will be lost.

Same thing happens in Narrowing Casting.

🎯 Key Points to Remember

✓ Type Casting converts one data type to another
✓ Two types → Widening & Narrowing
✓ Widening → Automatic & Safe
✓ Narrowing → Manual & Risk of data loss
✓ Used in calculations and memory control